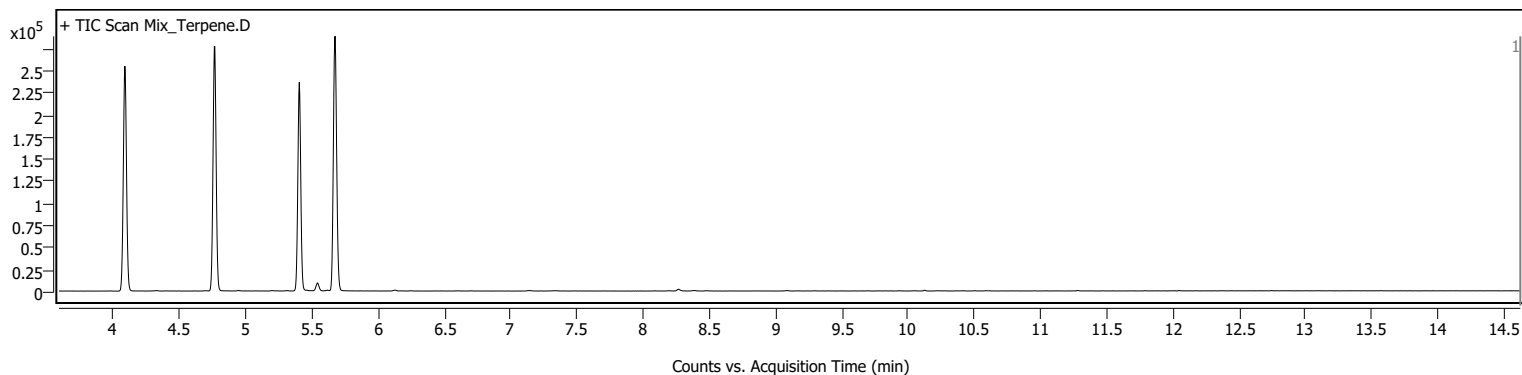
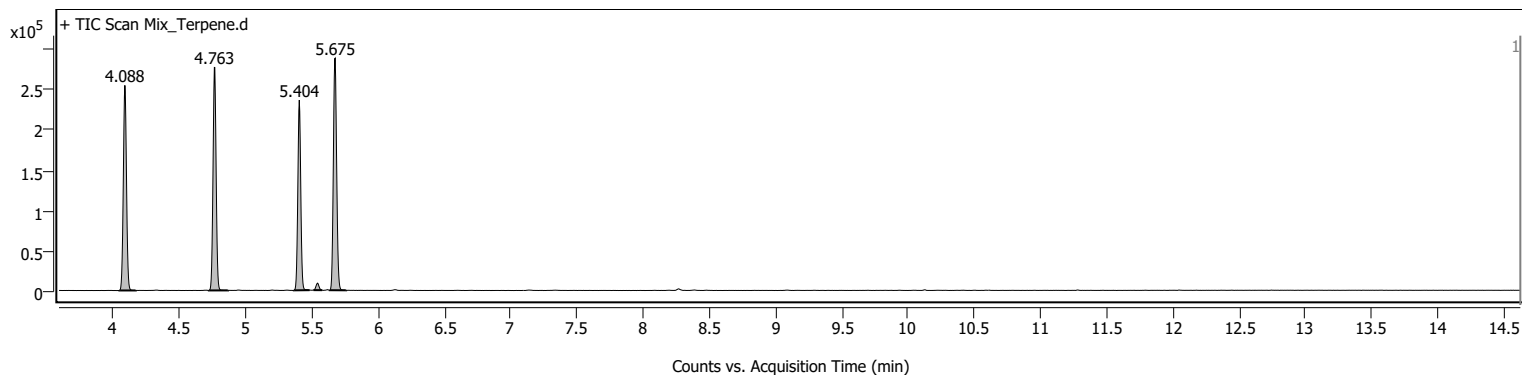
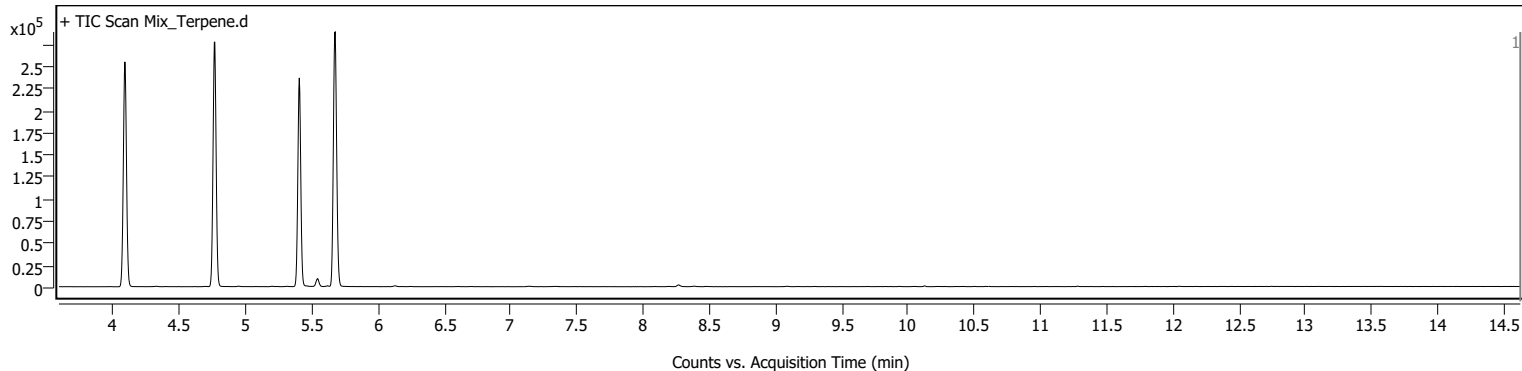


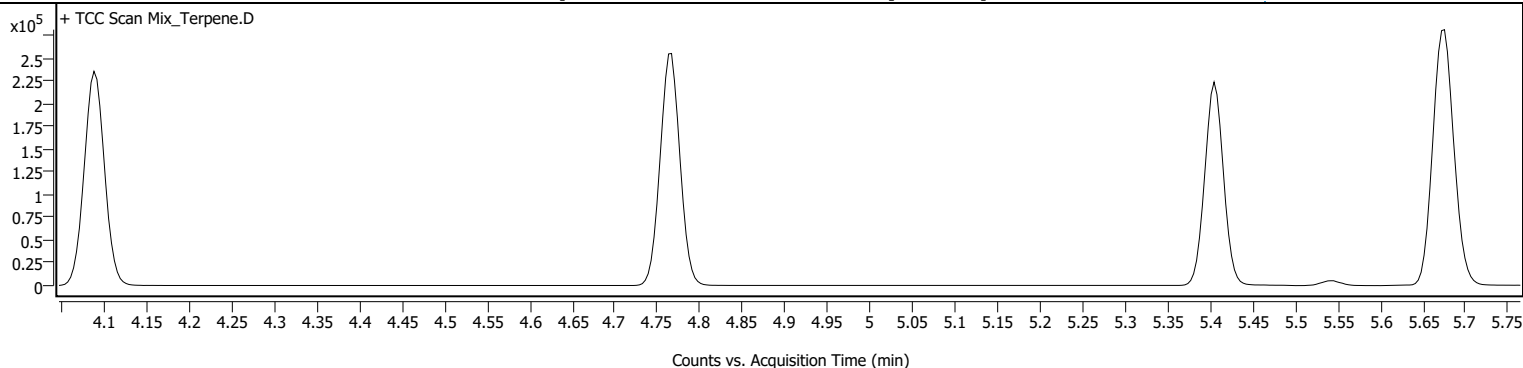
Sample Information

Name	Mix_Terpene	Data File Path	E:\MassHunter\GCMS\1\data\PraktikumWS25\Gruppe4\Mix_Terpene.D
Sample ID		Acq. Time (Local)	25/11/2025 15:30:31 (UTC+01:00)
Instrument	GCMSD	Method Path (Acq)	E:\MassHunter\GCMS\1\methods\Praktikum_Terpene_20split.M
MS Type	Q	Version (Acq SW)	MassHunter GC/MS Acquisition 10.2.530.3 21-Nov-2023 Copyright © 1989-2023 Agilent Technologies, Inc.
Inj. Vol. (ul)	1	IRM Status	
Position	2	Method Path (DA)	E:\MassHunter\Methods\10.0\Terpene_2024.m
Plate Pos.		Target Source Path	E:\MassHunter\Library\NIST17.L;E:\MassHunter\PCDL\default.csv
Operator		Result Summary	4 identified (5 found)

Sample Chromatograms



Compound Discovery Report



Compound Summary

Cpd	Name	Formula	RT	Mass	CAS	ID Source	Score	Score (Lib)	Score (DB)	Score (MFG)	Algorithm
1	.alpha.-Pinene	C10 H16	4.088		80-56-8	M-LibSearch	71.76	71.76			Chrom Decon
2	.beta.-Pinene	C10 H16	4.765		127-91-3	M-LibSearch	71.37	71.37			Chrom Decon
3	.gamma.-Terpinene	C10 H16	5.405		99-85-4	M-LibSearch	72.51	72.51			Chrom Decon
4			5.542								Chrom Decon
5	Eucalyptol	C10 H18 O	5.674		470-82-6	LibSearch	78.02	78.02			Chrom Decon

Compound Details

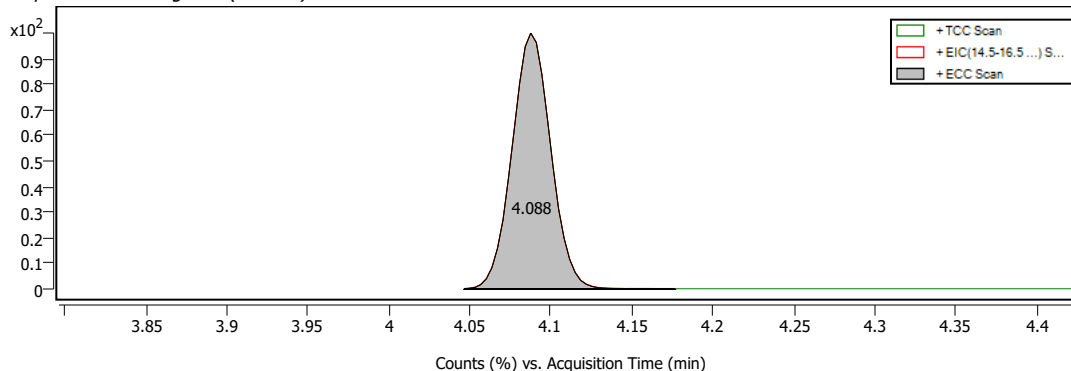
Cpd. 1: .alpha.-Pinene

Name	Formula	RT	RI	Mass	CAS	ID Source	Score	Algorithm	Lib/DB
.alpha.-Pinene	C10 H16	4.088			80-56-8	M-LibSearch	71.76	Chrom Decon	NIST17.L
Species	m/z	Score (Lib)	Num Spectra	Score (DB)	Hits	Score (MFG)	Score (RT)		
		71.76			15				

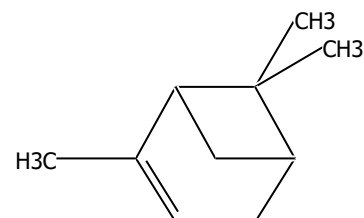
Compound ID Table

Name	Formula	Species	ID Source	RT	RT Diff	Mass	CAS	Score	Score (DB)	Score (MFG)
.alpha.-Pinene	C10H16		LibSearch	4.088	-11.712		80-56-8	71.76		
3,5-Methanocyclopentapyrazole, 3,3a,4,5,6,6a-hexahydro-3a,4,4-trimethyl-	C10H16N2		LibSearch	4.088			87143-58-6	88.62		
Bicyclo[3.1.1]hept-2-ene, 3,6,6-trimethyl-	C10H16		LibSearch	4.088	-11.712		4889-83-2	73.47		
1,4-Methano-1H-Cyclopropa[d]pyridazine, 4,4a,5,5a-tetrahydro-6,6-dimethyl-, (1.alpha.,4.alpha.,4a.alpha.,5a.alpha.)-	C8H12N2		LibSearch	4.088			109746-10-3	73.11		
Bicyclo[3.1.0]hex-2-ene, 2-methyl-5-(1-methylethyl)-	C10H16		LibSearch	4.088	-10.945		2867-05-2	72.67		
(1R)-2,6,6-Trimethylbicyclo[3.1.1]hept-2-ene	C10H16		LibSearch	4.088	-11.712		7785-70-8	72.44		
Tricyclo[2.2.1.0(2,6)]heptane, 1,3,3-trimethyl-	C10H16		LibSearch	4.088	-8.062		488-97-1	72.41		
Bicyclo[3.1.0]hex-2-ene, 4-methyl-1-(1-methylethyl)-	C10H16		LibSearch	4.088	-10.462		28634-89-1	72.39		
3-Carene	C10H16		LibSearch	4.088	-11.712		13466-78-9	72.28		
Tricyclo[2.2.1.0(2,6)]heptane, 1,3,3-trimethyl-	C10H16		LibSearch	4.088	-8.062		488-97-1	72.17		
Bicyclo[3.1.0]hex-2-ene, 2-methyl-5-(1-methylethyl)-	C10H16		LibSearch	4.088	-10.945		2867-05-2	71.89		
(1S)-2,6,6-Trimethylbicyclo[3.1.1]hept-2-ene	C10H16		LibSearch	4.088	-11.712		7785-26-4	71.69		
Cyclohexene, 4-methylene-1-(1-methylethyl)-	C10H16		LibSearch	4.088	-12.462		99-84-3	71.68		
Bicyclo[3.1.0]hex-2-ene, 2-methyl-5-(1-methylethyl)-	C10H16		LibSearch	4.088	-10.945		2867-05-2	70.94		
diazoadamantane	C10H14N2		LibSearch	4.088			24175-02-8	70.61		

Compound Chromatograms (overlaid)

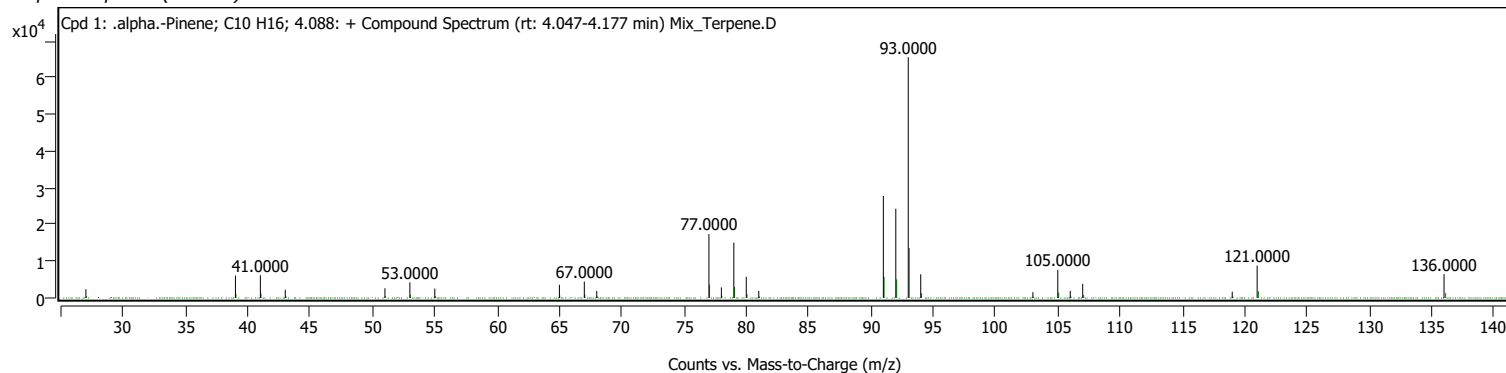


Structure



Compound Discovery Report

Compound Spectra (overlaid)



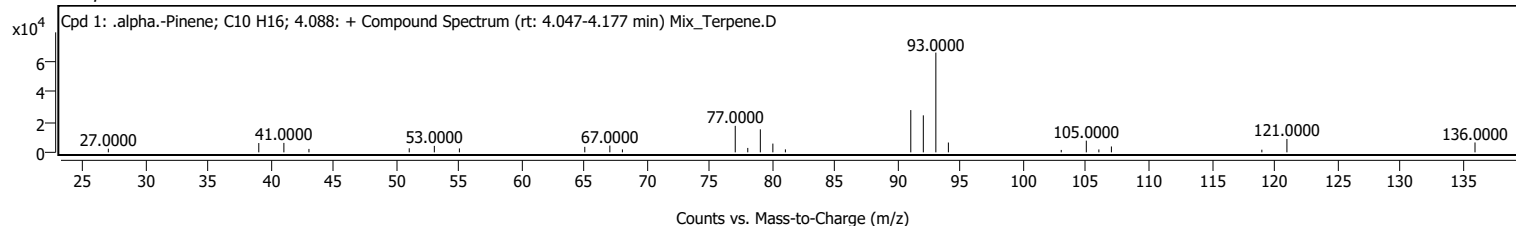
.alpha.-Pinene

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
			71.76			NIST17.L

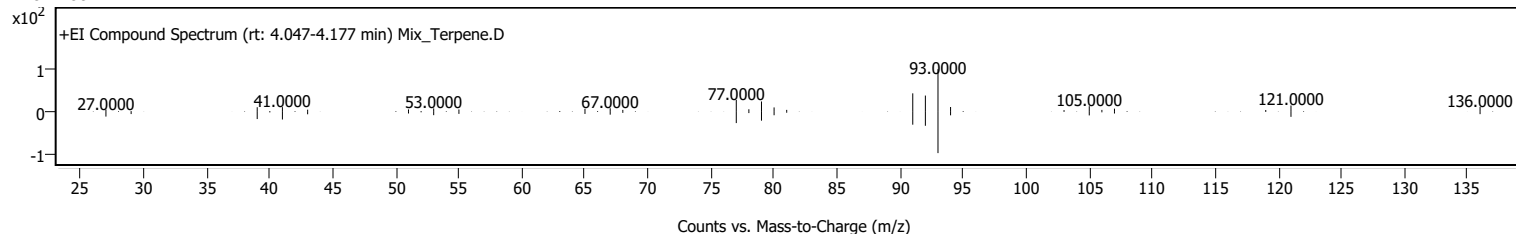
Library Match Details

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
			71.76			NIST17.L

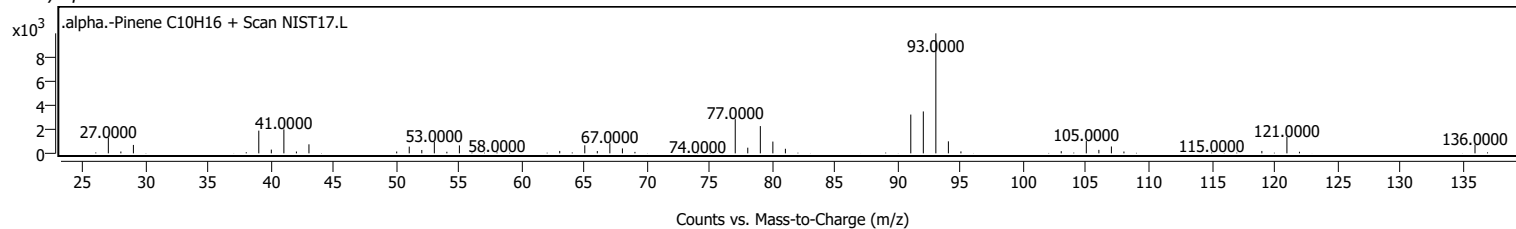
Observed Spectrum



Mirror Plot



Library Spectrum



Cpd. 2: .beta.-Pinene

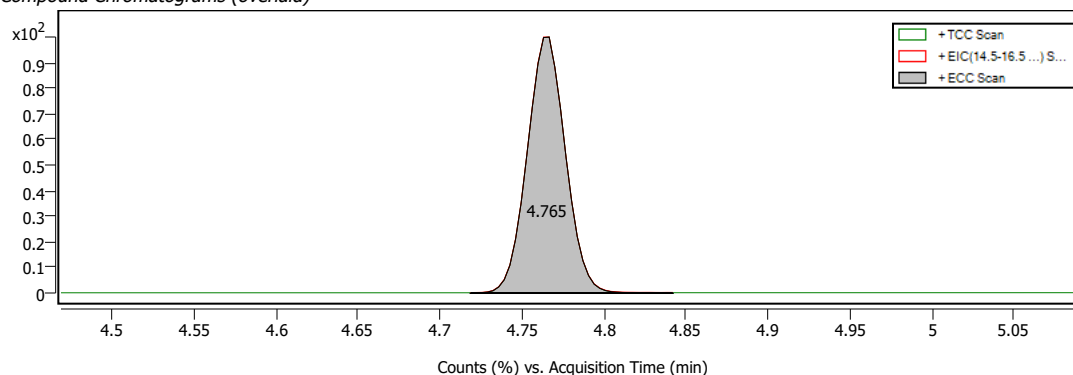
Name	Formula	RT	RI	Mass	CAS	ID Source	Score	Algorithm	Lib/DB
.beta.-Pinene	C10 H16	4.765			127-91-3	M-LibSearch	71.37	Chrom Decon	NIST17.L
Species	m/z	Score (Lib)	Num Spectra	Score (DB)	Hits	Score (MFG)	Score (RT)		
		71.37			15				

Compound Discovery Report

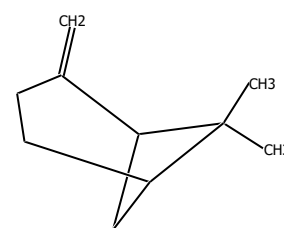
Compound ID Table

Name	Formula	Species	ID Source	RT	RT Diff	Mass	CAS	Score	Score (DB)	Score (MFG)
.beta.-Pinene	C10H16		LibSearch	4.765	-10.952		127-91-3	71.37		
3,5-Methanocyclopentapyrazole, 3,3a,4,5,6,6a-hexahydro-3a,4,4-trimethyl-	C10H16N2		LibSearch	4.765			87143-58-6	85.46		
Bicyclo[3.1.1]heptane, 6,6-dimethyl-2-methylene-, (1S)-	C10H16		LibSearch	4.765	-10.952		18172-67-3	75.25		
Spiro (6,6-dimethyl-2,3-diazobicyclo [3.1.0] hex-2-ene-4,1'-cyclopropane)	C8H12N2		LibSearch	4.765			135485-30-2	74.99		
Cyclohexene, 4-methylene-1-(1-methylethyl)-	C10H16		LibSearch	4.765	-11.785		99-84-3	74.95		
Bicyclo[3.1.0]hex-2-ene, 4-methyl-1-(1-methylethyl)-	C10H16		LibSearch	4.765	-9.785		28634-89-1	73.88		
Bicyclo[3.1.0]hexane, 4-methylene-1-(1-methylethyl)-	C10H16		LibSearch	4.765	-10.185		3387-41-5	73.74		
Bicyclo[3.1.0]hexane, 4-methylene-1-(1-methylethyl)-	C10H16		LibSearch	4.765	-10.185		3387-41-5	71.95		
Bicyclo[2.2.1]heptane, 7,7-dimethyl-2-methylene-	C10H16		LibSearch	4.765	-10.952		471-84-1	71.61		
Bicyclo[3.1.1]heptane, 6,6-dimethyl-2-methylene-, (1S)-	C10H16		LibSearch	4.765	-10.952		18172-67-3	71.29		
2,3-Diazabicyclo[2.2.1]hept-2-ene, 5-ethenyl-4,7,7-trimethyl-, (1.alpha.,4.alpha.,5.beta.)-	C10H16N2		LibSearch	4.765			1000221-84-7	71.12		
Spiro (6,6-dimethyl-2,3-diazobicyclo [3.1.0] hex-2-ene-4,1'-cyclopropane)	C8H12N2		LibSearch	4.765			135485-30-2	71.08		
Bicyclo[3.1.1]heptane, 6,6-dimethyl-2-methylene-, (1S)-	C10H16		LibSearch	4.765	-10.952		18172-67-3	70.93		
1,4-Methano-1H-Cyclopropa[d]pyridazine, 4,4a,5,5a-tetrahydro-6,6-dimethyl-, (1.alpha.,4.alpha.,4a.alpha.,5a.alpha.)-	C8H12N2		LibSearch	4.765			109746-10-3	70.85		
.beta.-Phellandrene	C10H16		LibSearch	4.765	-11.302		555-10-2	70.81		

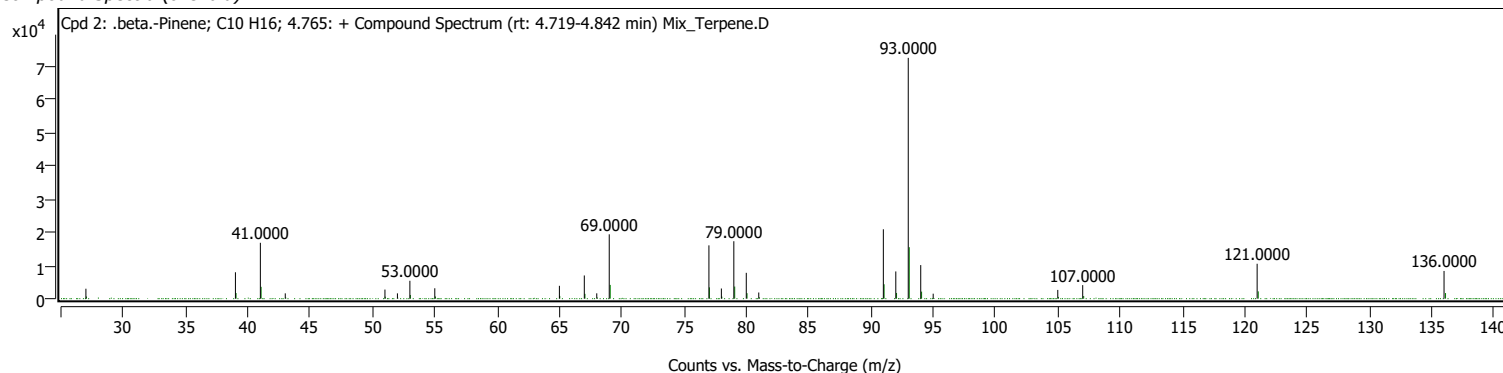
Compound Chromatograms (overlaid)



Structure



Compound Spectra (overlaid)



.beta.-Pinene

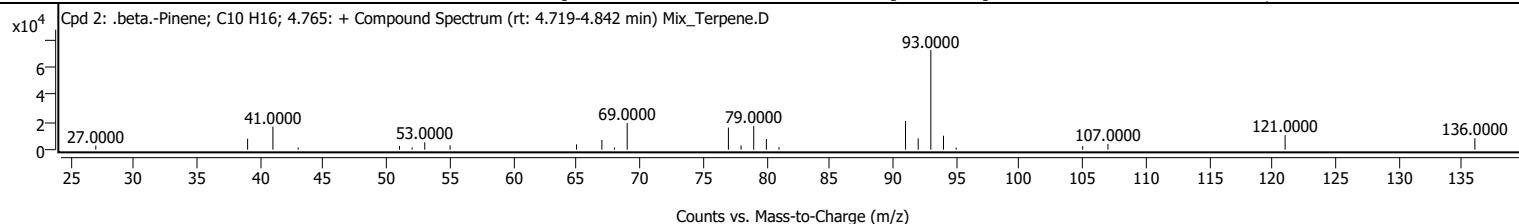
m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
71.37						NIST17.L

Library Match Details

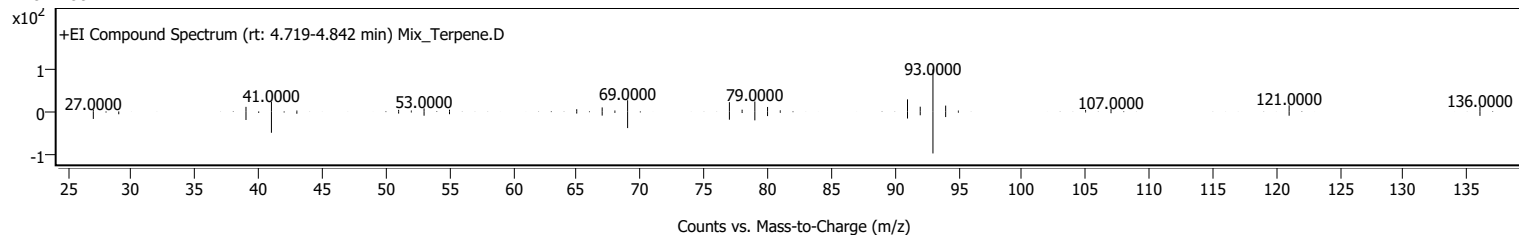
m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
71.37						NIST17.L

Observed Spectrum

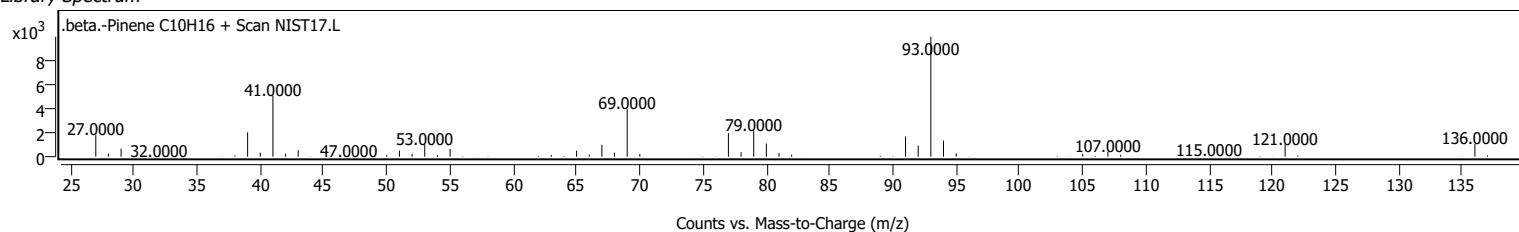
Compound Discovery Report



Mirror Plot



Library Spectrum



Cpd. 3: .gamma.-Terpinene

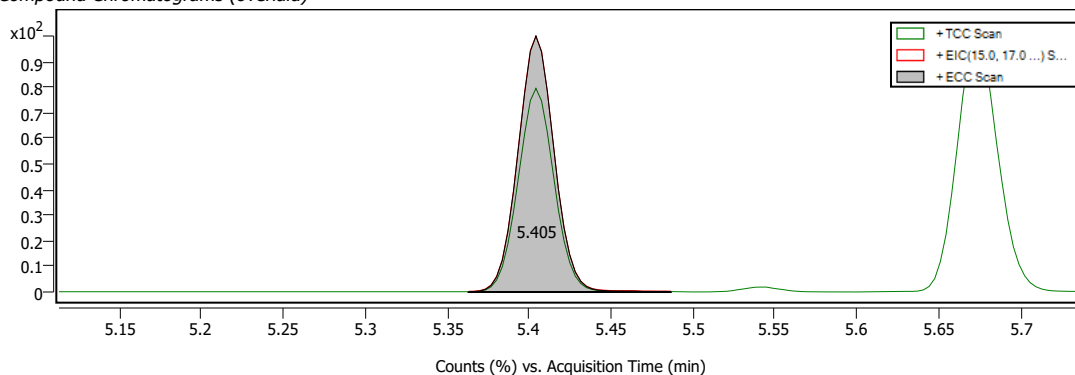
Name	Formula	RT	RI	Mass	CAS	ID Source	Score	Algorithm	Lib/DB
.gamma.-Terpinene	C10 H16	5.405			99-85-4	M-LibSearch	72.51	Chrom Decon	NIST17.L
Species	m/z	Score (Lib)	Num Spectra	Score (DB)	Hits	Score (MFG)	Score (RT)		
		72.51			15				

Compound ID Table

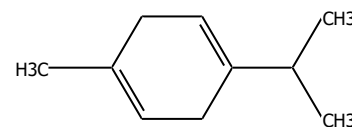
Name	Formula	Species	ID Source	RT	RT Diff	Mass	CAS	Score	Score (DB)	Score (MFG)
.gamma.-Terpinene	C10H16		LibSearch	5.405	-11.229		99-85-4	72.51		
Cyclopropane, trimethyl(2-methyl-1-propenylidene)-	C10H16		LibSearch	5.405			14803-30-6	80.83		
1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	C10H16		LibSearch	5.405	-11.229		99-86-5	75.90		
(+)-4-Carene	C10H16		LibSearch	5.405	-9.912		29050-33-7	73.40		
Cyclohexene, 1-methyl-4-(1-methylethylidene)-	C10H16		LibSearch	5.405	-12.129		586-62-9	72.93		
1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	C10H16		LibSearch	5.405	-11.229		99-86-5	72.86		
Cyclohexene, 1-methyl-4-(1-methylethylidene)-	C10H16		LibSearch	5.405	-12.129		586-62-9	72.65		
3,5-Methanocyclopentapyrazole, 3,3a,4,5,6,6a-hexahydro-3a,4,4-trimethyl-	C10H16N2		LibSearch	5.405			87143-58-6	72.48		
1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	C10H16		LibSearch	5.405	-11.229		99-86-5	72.31		
Cyclohexene, 1-methyl-4-(1-methylethylidene)-	C10H16		LibSearch	5.405	-12.129		586-62-9	72.09		
Bicyclo[2.2.1]hept-2-ene, 1,7,7-trimethyl-	C10H16		LibSearch	5.405	-10.129		464-17-5	72.04		
1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	C10H16		LibSearch	5.405	-11.229		99-86-5	72.03		
.gamma.-Terpinene	C10H16		LibSearch	5.405	-11.229		99-85-4	71.97		
Bicyclo[3.1.0]hex-2-ene, 4,4,6,6-tetramethyl-	C10H16		LibSearch	5.405	-9.145		19487-09-3	71.90		
Cyclohexene, 3-methyl-6-(1-methylethylidene)-	C10H16		LibSearch	5.405	-11.645		586-63-0	71.48		

Compound Discovery Report

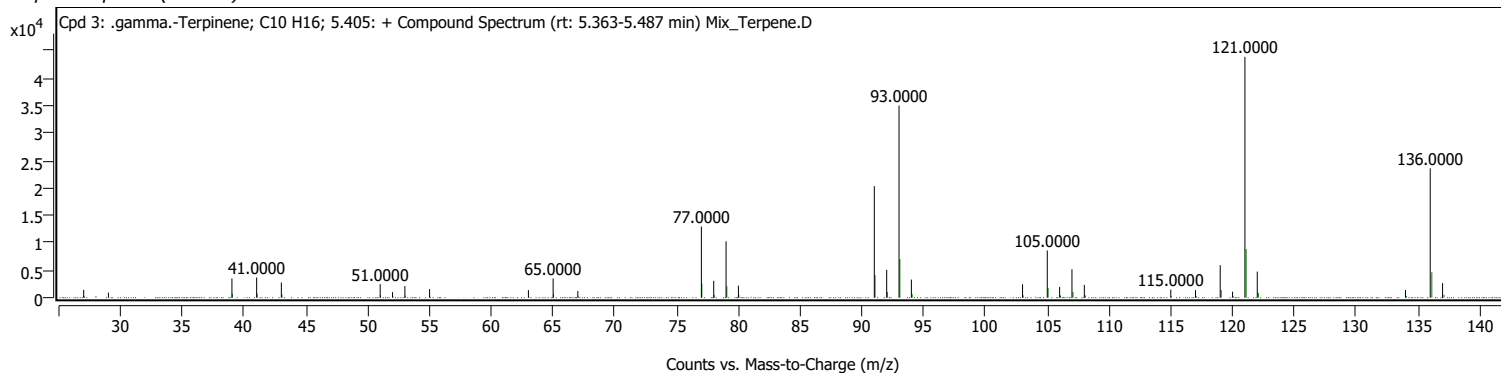
Compound Chromatograms (overlaid)



Structure



Compound Spectra (overlaid)



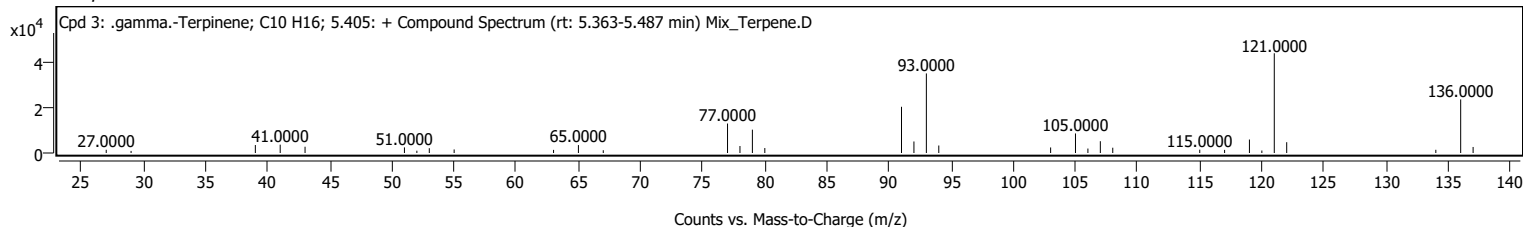
.gamma.-Terpinene

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
72.51						NIST17.L

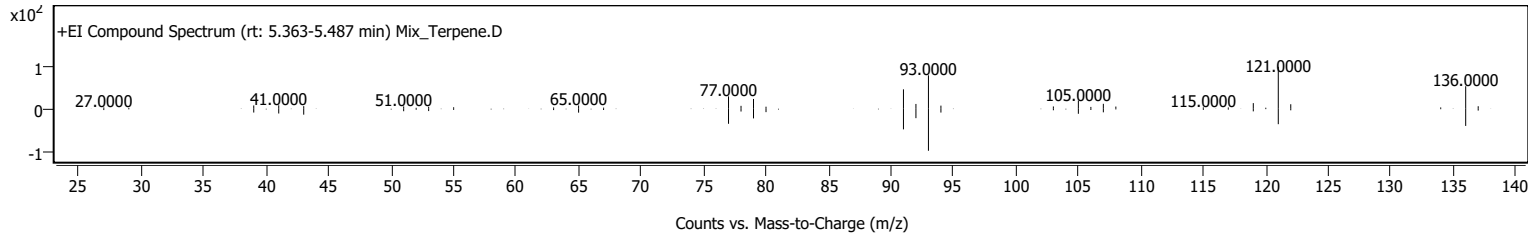
Library Match Details

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
72.51						NIST17.L

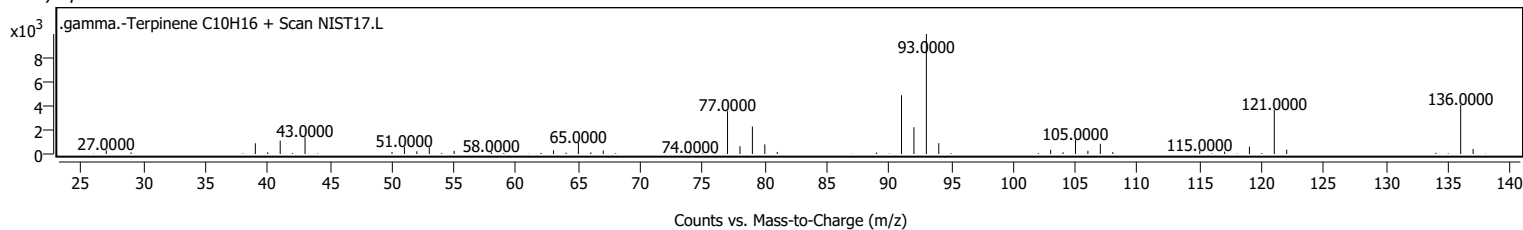
Observed Spectrum



Mirror Plot



Library Spectrum

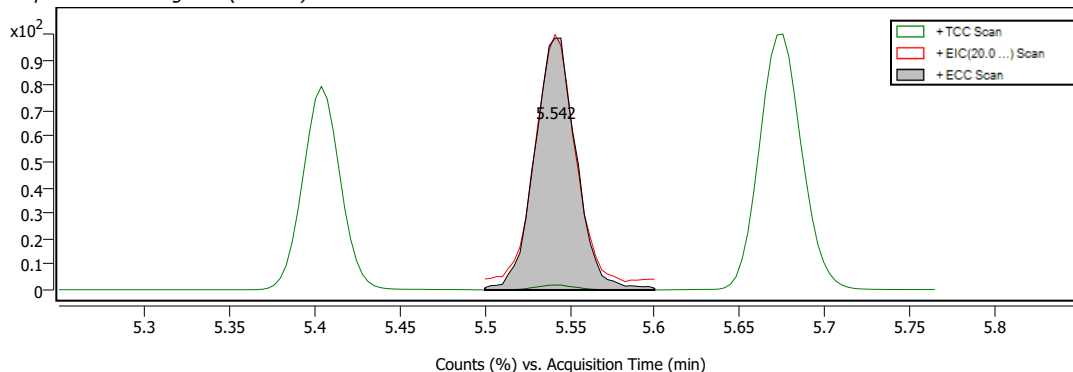


Cpd 4: 5.542

Compound Discovery Report

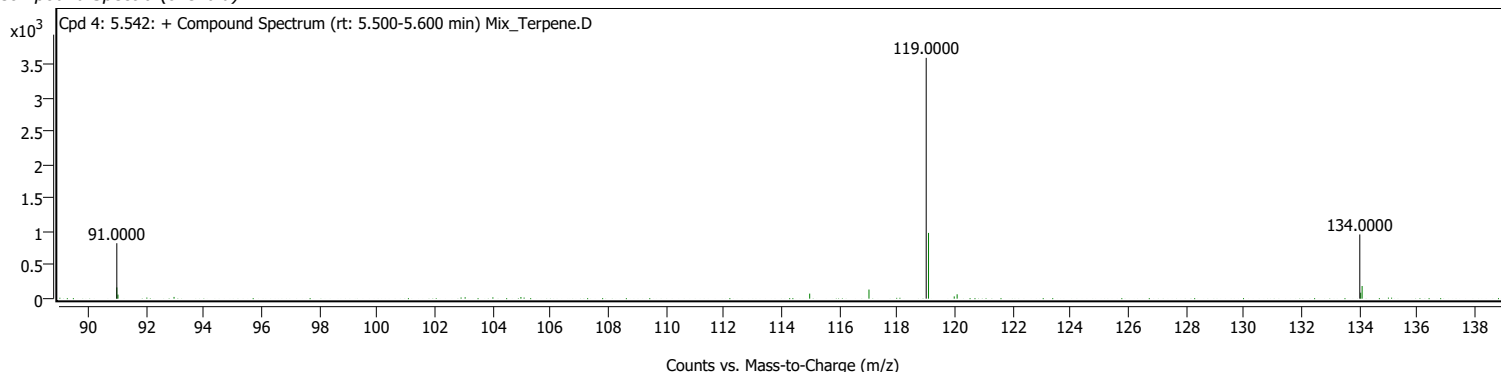
Name	Formula	RT	RI	Mass	CAS	ID Source	Score	Algorithm	Lib/DB
		5.542						Chrom Decon	

Compound Chromatograms (overlaid)



Structure

Compound Spectra (overlaid)



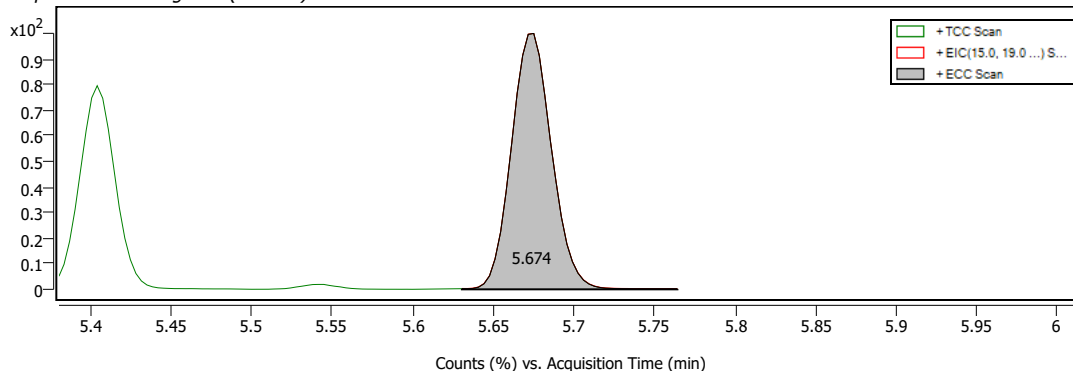
Cpd. 5: Eucalyptol

Name	Formula	RT	RI	Mass	CAS	ID Source	Score	Algorithm	Lib/DB
Eucalyptol	C10 H18 O	5.674			470-82-6	LibSearch	78.02	Chrom Decon	NIST17.L
Species	m/z	Score (Lib)	Num Spectra	Score (DB)	Hits	Score (MFG)	Score (RT)		
		78.02			5				

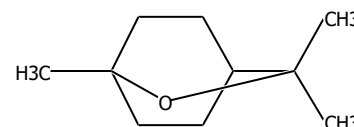
Compound ID Table

Name	Formula	Species	ID Source	RT	RT Diff	Mass	CAS	Score	Score (DB)	Score (MFG)
Eucalyptol	C10H18O		LibSearch	5.674	-11.976		470-82-6	78.02		
Eucalyptol	C10H18O		LibSearch	5.674	-11.976		470-82-6	76.94		
Eucalyptol	C10H18O		LibSearch	5.674	-11.976		470-82-6	76.85		
Eucalyptol	C10H18O		LibSearch	5.674	-11.976		470-82-6	75.56		
Eucalyptol	C10H18O		LibSearch	5.674	-11.976		470-82-6	72.82		

Compound Chromatograms (overlaid)

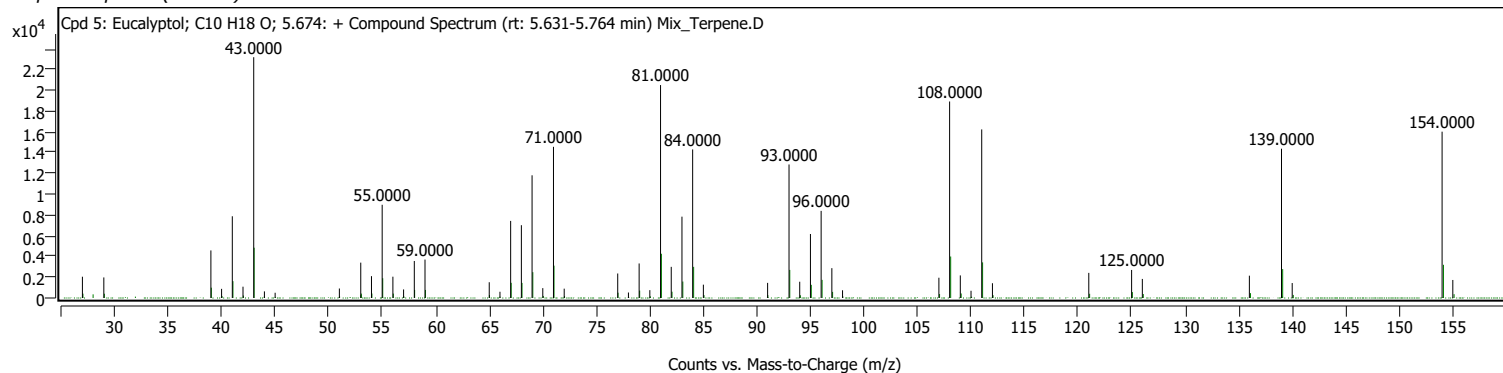


Structure



Compound Discovery Report

Compound Spectra (overlaid)



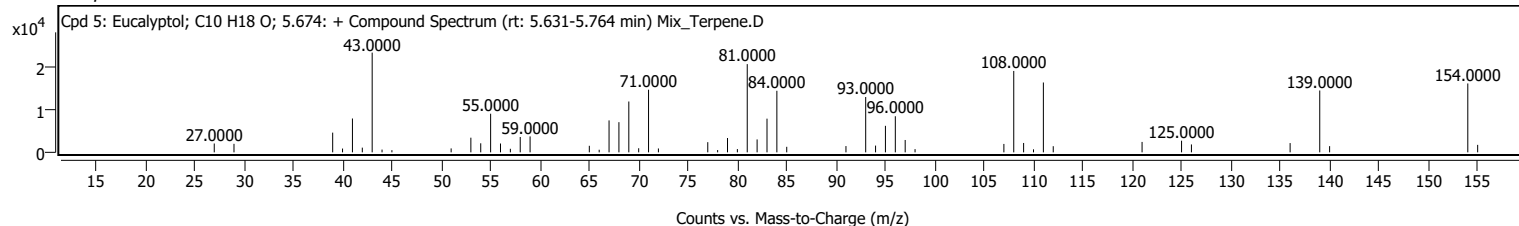
Eucalyptol

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
			78.02			NIST17.L

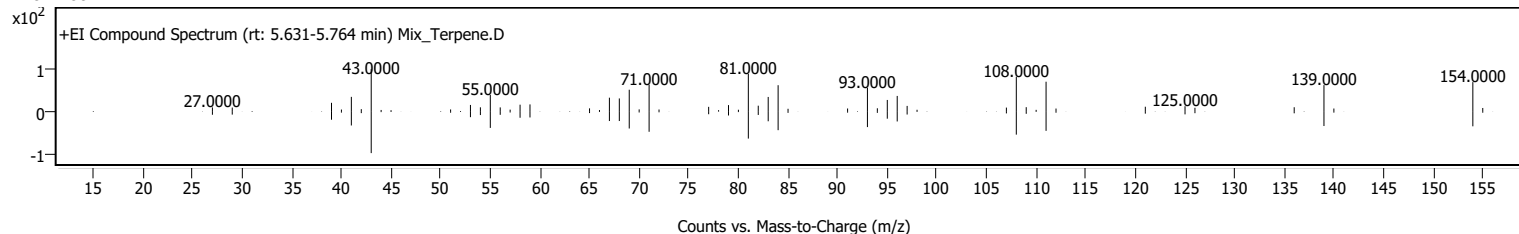
Library Match Details

m/z (prec.)	Species	CE	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
			78.02			NIST17.L

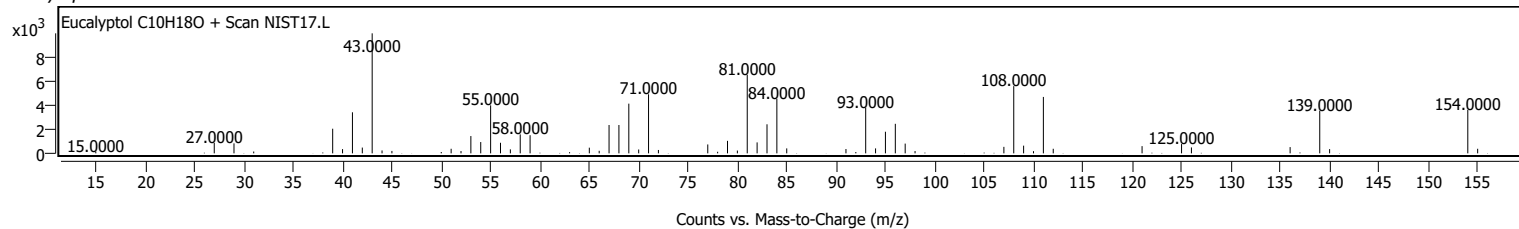
Observed Spectrum



Mirror Plot



Library Spectrum



MassHunter Qual 10.0
(End of Report)